

Text File Handling in Java

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Text File handling is a process of storing and retrieving data from a text file(.txt) using Java programs.

A text file stores data in a human-readable format.

Examples:

student.txt
marks.txt
poem.txt
story.txt

Java provides several classes for file handling:

- File
- FileWriter
- FileReader
- BufferedWriter
- BufferedReader
- PrintWriter
- Scanner

All these classes belongs to the java.io package (except Scanner which belongs to java.util package)

Why Use Files?

Variable store data temporarily in memory. When the program terminates data is lost.

Files provide:

- Permanent storage
- Data sharing
- Data backup
- Record maintenance

File Class

The File class is used to create, delete and manage files.
(We already studied about it.)

Writing Data to a Text File

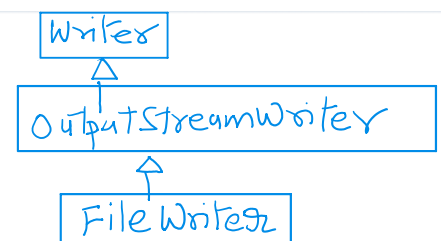
Using FileWriter class

Class FileWriter

java.lang.Object
java.io.Writer
java.io.OutputStreamWriter
java.io.FileWriter

All Implemented Interfaces:

Closeable, Flushable, Appendable, AutoCloseable



Class FileWriter

```
java.lang.Object
  java.io.Writer
    java.io.OutputStreamWriter
      java.io.FileWriter
```

All Implemented Interfaces:

Closeable, Flushable, Appendable, AutoCloseable

```
public class FileWriter
extends OutputStreamWriter
```

Writes text to character files using a default buffer size. Encoding from characters to bytes uses either a specified `charset` or the default `charset`.

Whether or not a file is available or may be created depends upon the underlying platform. Some platforms, in particular, allow a file to be opened for writing by only one `FileWriter` (or other file-writing object) at a time. In such situations the constructors in this class will fail if the file involved is already open.

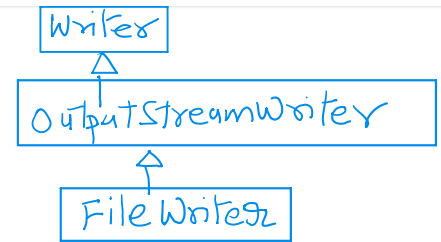
The `FileWriter` is meant for writing streams of characters. For writing streams of raw bytes, consider using a `FileOutputStream`.

Since:

1.1

See Also:

`OutputStreamWriter`, `FileOutputStream`, `Charset.defaultCharset()`



Constructors

Constructor	Description
✓ <code>FileWriter(File file)</code>	Constructs a <code>FileWriter</code> given the <code>File</code> to write, using the default <code>charset</code> .
<code>FileWriter(FileDescriptor fd)</code>	Constructs a <code>FileWriter</code> given a file descriptor, using the default <code>charset</code> .
✓ <code>FileWriter(File file, boolean append)</code>	Constructs a <code>FileWriter</code> given the <code>File</code> to write and a boolean indicating whether to append the data written, using the default <code>charset</code> .
✓ <code>FileWriter(File file, Charset charset)</code>	Constructs a <code>FileWriter</code> given the <code>File</code> to write and <code>charset</code> .
<code>FileWriter(File file, Charset charset, boolean append)</code>	Constructs a <code>FileWriter</code> given the <code>File</code> to write, <code>charset</code> and a boolean indicating whether to append the data written.
✓ <code>FileWriter(String fileName)</code>	Constructs a <code>FileWriter</code> given a file name, using the default <code>charset</code> .
✓ <code>FileWriter(String fileName, boolean append)</code>	Constructs a <code>FileWriter</code> given a file name and a boolean indicating whether to append the data written, using the default <code>charset</code> .
<code>FileWriter(String fileName, Charset charset)</code>	Constructs a <code>FileWriter</code> given a file name and <code>charset</code> .
<code>FileWriter(String fileName, Charset charset, boolean append)</code>	Constructs a <code>FileWriter</code> given a file name, <code>charset</code> and a boolean indicating whether to append the data written.

```
package FileHandling;
import java.io.*;
import java.util.*;

/**
 * Write a description of class WriteFile here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class WriteFile
{
    public static void main(String[] args){
```

```

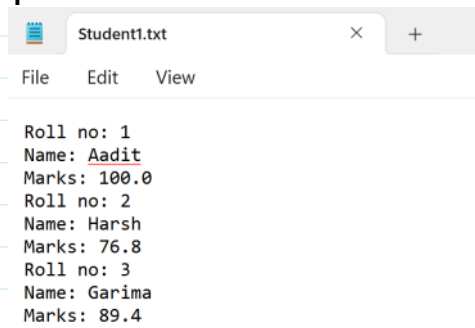
Scanner sc = new Scanner(System.in);
String fileName, path = "./DataFiles/";
System.out.print("\nEnter file name with .txt extension: ");
fileName = sc.nextLine();
try{
    FileWriter fw = new FileWriter(path+fileName, true);
    System.out.print("\nEnter Roll number: ");
    int rollNum = sc.nextInt();
    sc.nextLine(); //clears the keyboard buffer
    System.out.print("Enter name: ");
    String name = sc.nextLine();
    System.out.print("Enter marks: ");
    float marks = sc.nextFloat();
    fw.write("Roll no: " + rollNum+"\n");
    fw.write("Name: " + name + "\n");
    fw.write("Marks: " + marks + "\n");

    fw.close();
    System.out.print("\nData Written");
}catch(IOException ioe){
    System.err.print("\nFile is not created");
}

}
}

```

Output:



Reading Data from a Text File Using FileReader

```

package FileHandling;
import java.util.*;
import java.io.*;

/**
 * Write a description of class ReadFile here.
 *
 * @author (your name)
 * @version (a version number or a date)

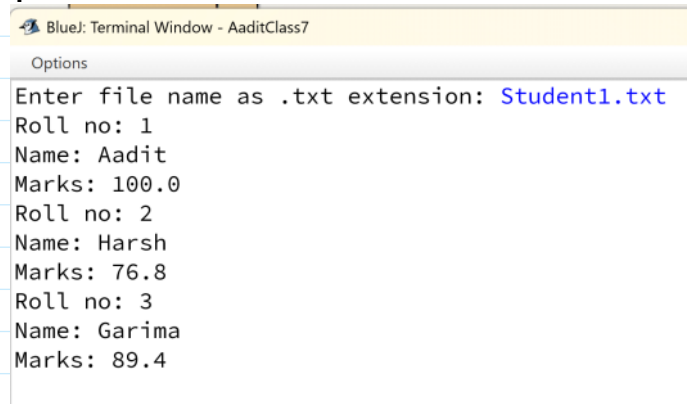
```

```

*/
public class ReadFile
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String path="./DataFiles/", fileName;
        System.out.print("\nEnter file name as .txt extension: ");
        fileName = sc.nextLine();
        try{
            FileReader fr = new FileReader(path+fileName);
            int ch;
            while((ch = fr.read())!= -1){
                System.out.print((char)ch);
            }
            fr.close();
        }catch(FileNotFoundException fnfe){
            System.err.print("\nFile nahi mila.");
        }
        catch(IOException ioe){
            System.err.print("\nNetwork Error!");
        }
    }
}

```

Output:



```

BlueJ: Terminal Window - AaditClass7
Options
Enter file name as .txt extension: Student1.txt
Roll no: 1
Name: Aadit
Marks: 100.0
Roll no: 2
Name: Harsh
Marks: 76.8
Roll no: 3
Name: Garima
Marks: 89.4

```

Using BufferedWriter

BufferedWriter improves writing performance.

```
package FileHandling;
```

```
import java.util.*;
```

```
import java.io.*;
```

```
/**
```

```
 * Write a description of class UsingBufferedWriter here.
```

```
 *
```

```
 * @author (your name)
```

```
 * @version (a version number or a date)
```

```
 */
```

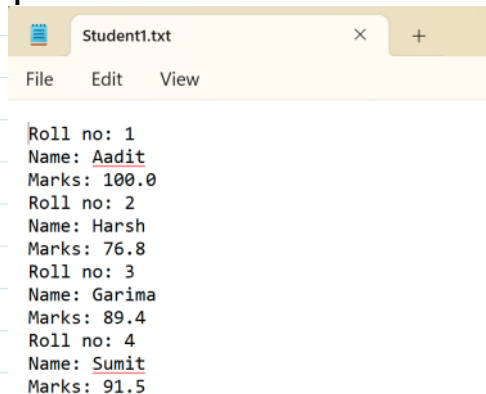
```

public class UsingBufferedWriter
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String fileName, path = "./DataFiles/";
        System.out.print("\nEnter file name with .txt extension: ");
        fileName = sc.nextLine();
        try{
            FileWriter fw = new FileWriter(path+fileName, true);
            BufferedWriter bw = new BufferedWriter(fw);
            System.out.print("\nEnter Roll number: ");
            int rollNum = sc.nextInt();
            sc.nextLine(); //clears the keyboard buffer
            System.out.print("Enter name: ");
            String name = sc.nextLine();
            System.out.print("Enter marks: ");
            float marks = sc.nextFloat();
            bw.write("Roll no: " + rollNum);
            bw.newLine();
            bw.write("Name: " + name);
            bw.newLine();
            bw.write("Marks: " + marks);
            bw.newLine();

            bw.close();
            System.out.print("\nData Written");
        }catch(IOException ioe){
            System.err.print("\nFile is not created");
        }
    }
}

```

Output:



```

Student1.txt
File Edit View
Roll no: 1
Name: Aadit
Marks: 100.0
Roll no: 2
Name: Harsh
Marks: 76.8
Roll no: 3
Name: Garima
Marks: 89.4
Roll no: 4
Name: Sumit
Marks: 91.5

```